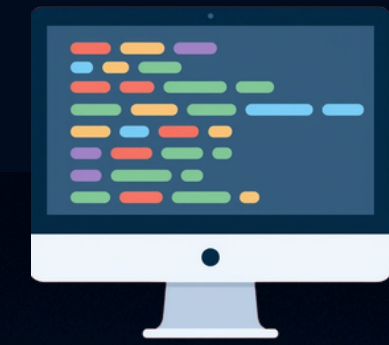


<Fundamentals of Programming>

<Programming Language: C>



INVENTORY MANAGEMENT



Presented by:

Ananya Salvi – 2962

Arya Rao – 3969

Mannat Parwani – 4086

Purva Sharma – 3538

Tanvi Wadekar – 2997

Presented to:

Mr. Nikhil
Soniminde

Group_Alpha
Group_Alpha

Username:

Password:

Login

Authorized Users:

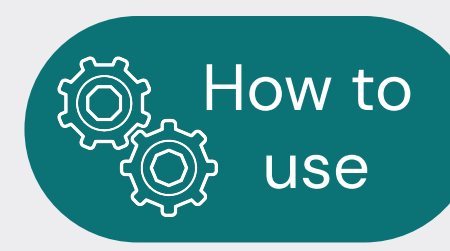
Ananya

Mannat

Purva

Arya

Tanvi



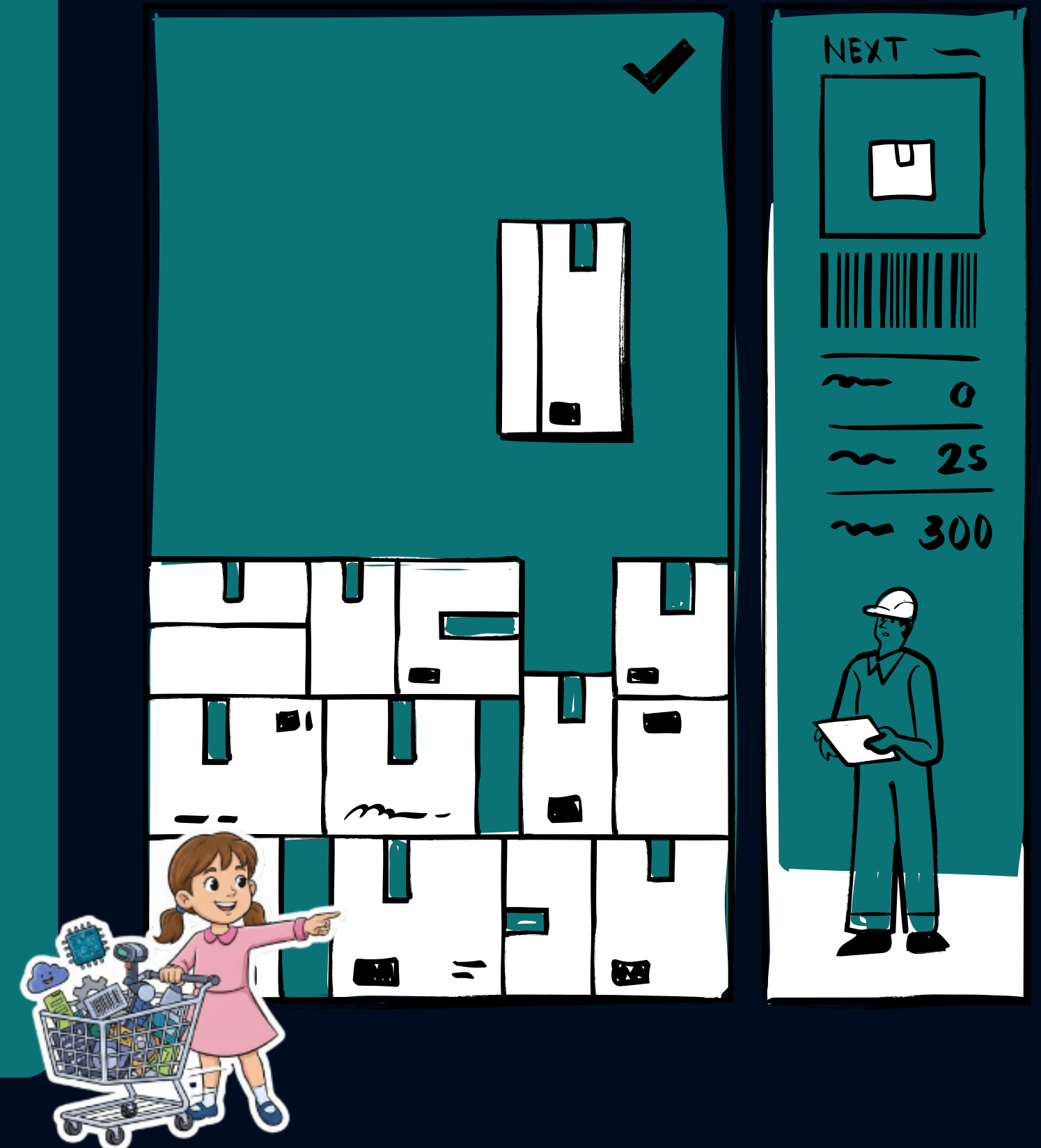
Problem Statement:

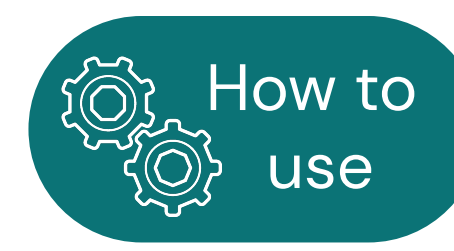
CHALLENGES IN TRADITIONAL INVENTORY MANAGEMENT

What is Inventory Management?

Inventory management is the process of tracking, storing, and controlling goods and materials in a business to ensure the right products are available at the right time.

- Manual inventory is **time-consuming** and **error-prone**.
- Poor tracking systems cause **stock mismanagement**.
- Businesses need a **simple digital solution**.





Objectives:

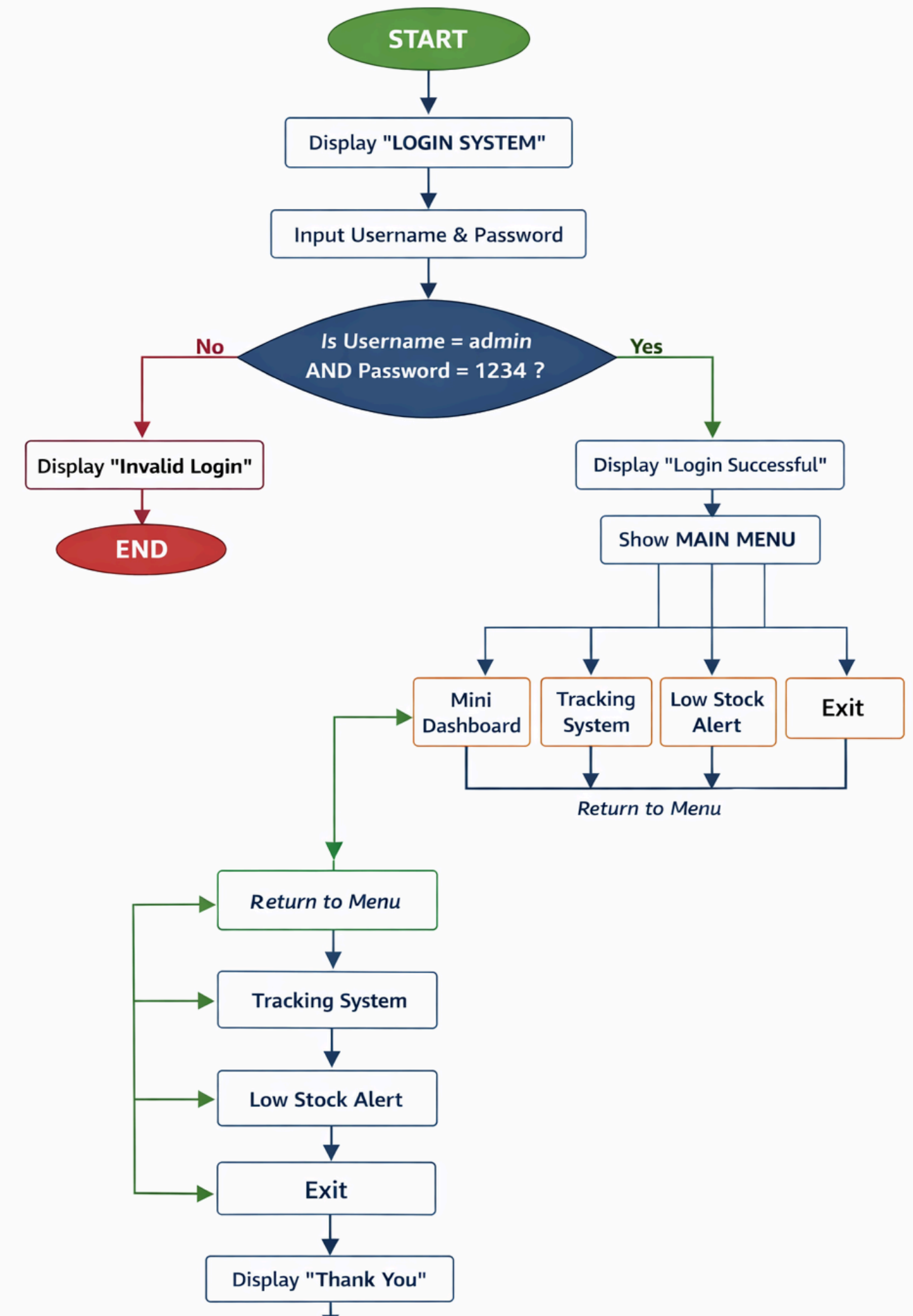
WHAT OUR SYSTEM AIMS TO ACHIEVE

- **Digitize** manual inventory processes
- Provide **real-time** stock updates
- Reduce **human errors**
- Enable **data-driven** decision making
- Improve productivity and **save time**



FLOWCHART

SYSTEM WORKING OVERVIEW



TECH STACK

Programming Language

C Programming Language



Core Concepts Used

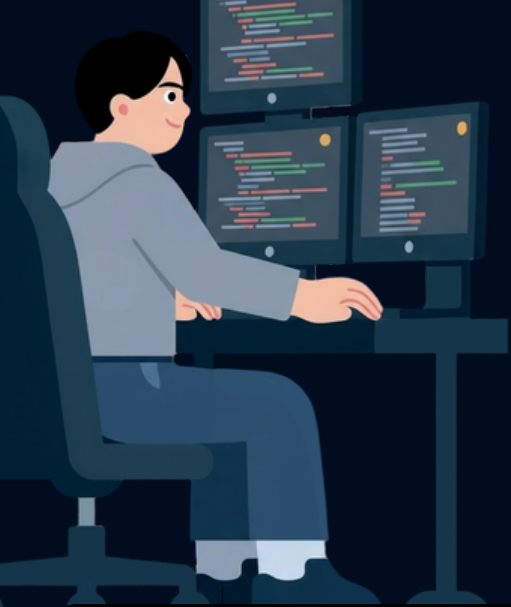
- 1) Arrays
- 2) Structures
- 3) Functions
- 4) Conditional Statements
- 5) Loops
- 6) String Handling (string.h)
- 7) Basic Authentication Logic

Libraries Used

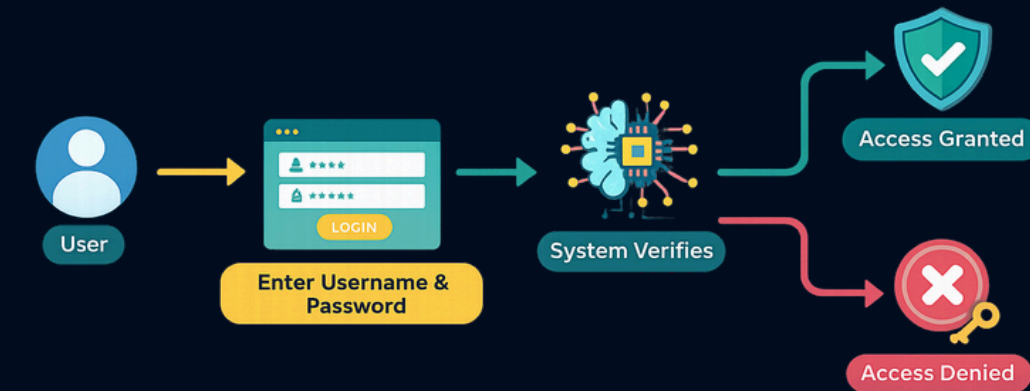
stdio.h (Input/Output functions)
string.h (String comparison)

Development Environment

VS Code



How to use (Behind the scenes)



⋮

< > 🔍 Login Code ↻

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char u[20], p[20];
6     printf(" LOGIN SYSTEM\n");
7     printf("Username: ");
8     scanf("%s", u);
9
10    printf("Password: ");
11    scanf("%s", p);
12
13    if(strcmp(u,"admin")==0 && strcmp(p,"1234")==0)
14        printf("Login Successful");
15    else
16        printf("Invalid Login");
17
18    return 0;
19 }
20
```

⋮

Output

```
LOGIN SYSTEM
Username: Password: Login Successful
[Execution complete with exit code 0]
```

⋮

Output

```
MINI DASHBOARD
Total Items      : 20
Total Stock      : 678
Low Stock Alerts : 0
```

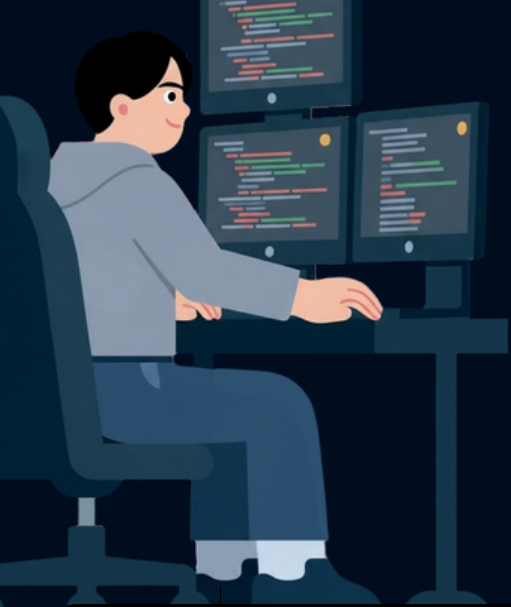
⋮

< > 🔍 Mini Dashboard ↻

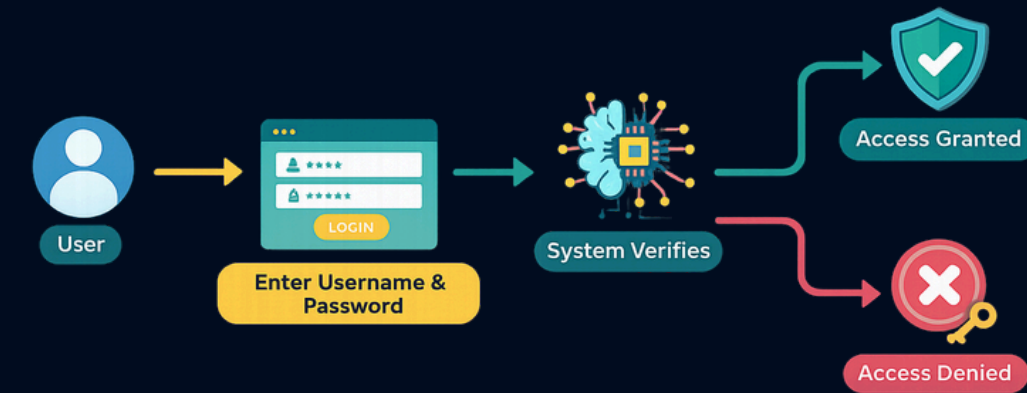
```
1 #include<stdio.h>
2 int main()
3 {
4     int quantity[20] = {40,35,25,20,50,18,12,60,45,22,15,30,18,55,28,26,20,14,61,40};
5     int threshold[20] = {15,15,10,8,20,10,10,25,20,10,8,15,10,25,10,10,10,8,30,40};
6
7     int totalItems = 20;
8     int totalStock = 0;
9     int lowStockItems = 0;
10
11    for(int i = 0; i < totalItems; i++)
12    {
13        totalStock += quantity[i];
14        if(quantity[i] <= threshold[i])
15        {
16            lowStockItems++;
17        }
18    }
19    printf(" MINI DASHBOARD\n");
20    printf("Total Items      : %d\n", totalItems);
21    printf("Total Stock       : %d\n", totalStock);
22    printf("Low Stock Alerts: %d\n", lowStockItems);
23    return 0;
24 }
25
```

Used in: ATMs • Websites • Apps • Secure Systems

Analyzes stock and displays system summary.



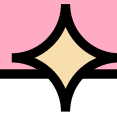
How to use (Behind the scenes)



ooo

< > 🔍 Tracking System Code ↻

```
1 #include <stdio.h>
2 int main()
3 {
4     int id[5] = {101,102,103,104,105};
5     int quantity[5] = {40,35,25,20,50};
6     float price[5] = {420,260,180,95,45};
7     printf("TRACKING SYSTEM\n");
8     int search, i;
9     printf("Enter Item ID: ");
10    scanf("%d", &search);
11    for(i = 0; i < 5; i++)
12    {
13        if(search==id[i])
14        {
15            printf("Item Found\n");
16            printf("Quantity: %d\n", quantity[i]);
17            printf("Price: %.2f\n", price[i]);
18            return 0;
19        }
20    }
21    printf("Item not found");
22    return 0;
23 }
24
```



Output

```
TRACKING SYSTEM
Enter Item ID: Item Found
Quantity: 35
Price: 260.00
```

[Execution complete with exit code 0]



Output

```
LOW STOCK ALERT
Item ID 103 is LOW in stock (Qty: 8)
Item ID 104 is LOW in stock (Qty: 5)
```

[Execution complete with exit code 0]

ooo

< > 🔍 Low Stock Alert Code ↻

```
1 #include <stdio.h>
2 struct Item
3 {
4     int id;
5     int quantity;
6     int threshold;
7 };
8 int main()
9 {
10     struct Item items[5] =
11     {
12         {101,40,15},
13         {102,35,15},
14         {103,8,10},
15         {104,5,8},
16         {105,50,20}
17     };
18     int i;
19     printf("LOW STOCK ALERT\n");
20     for(i = 0; i < 5; i++) {
21         if(items[i].quantity <= items[i].threshold)
22         {
23             printf("Item ID %d is LOW in stock (Qty: %d)\n",
24                 items[i].id, items[i].quantity);
25         }
26     }
27     return 0;
28 }
29
```

Find the item. Know the stock. Stay in control.

Detect shortages before they become problems.

The "Boss Level" Challenges (THE LIMITATIONS)



**AISE BADE BADE MINI
PROJECT ME GHHOTI
GHHOTI LIMITATION
TOH HOTI HI HAI**

- **No Cloud Storage:**
"Data is local only—no cloud, no worries (for now)!"
- **No Multi-User:**
"Single Admin mode: Only one captain on this ship."
- **No Auto-Billing:**
"Manual entry for sales—keeps your math skills sharp!"
- **Small Scale Only:**
"Optimized for small vendors—Enterprise mode is still loading!"

Advanced C Programmmion

Advanced Programming Concepts

Pointers

Memory Management

Bit Manipulation

Data Structures

Linked Lists

Function Pointers

Bit Manipulation

Function Pointers

Bit Manipulation



Future Scope

Multilingual Support

दिएसपहरी करानी | .argentina laap pa.gels

द्वैमला English | →

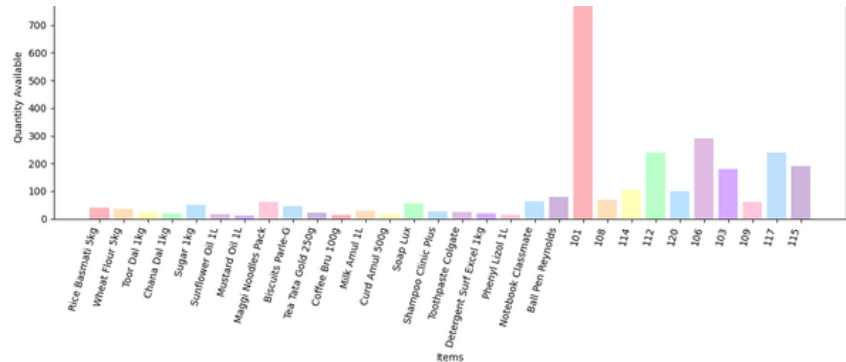
जंस्वर English मेयस

देवादाव शदुरं

वौद्वित थो डो मल्लिचमत्
करिस्नोत् कृत्यहो जान
बुफुयर कर्सि चैं हो कोत्



Advanced Data Visualization & Analytics



Intelligent Auto Suggestion System

automatically suggests reorder quantities based on past sales patterns and stock movement trends.

This will reduce manual monitoring and improve decision-making efficiency



Multilingual Interface Support

The system can be upgraded to support multiple regional languages such as Hindi, Marathi, and English, making it accessible to small business owners across different regions of India.

Advanced C Concepts

- File Handling in C (Using fopen(), fclose(), fprintf(), fscanf())
- Dynamic Memory Allocation (Using malloc(), calloc(), free())
- Linked List Data Structure
- Binary File Management System
- Structure with Pointers
- Sorting Algorithms (Quick Sort / Merge Sort)

THANK YOU

Any questions, feel free to ask!
All questions are welcome... unless
they're out of syllabus!

